Xuanzhuo Liu

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EDUCATION

The Chinese University of Hong Kong, Shenzhen Shenzhen, China
Bachelor of Science, Data Science and Big Data Technology (Advisor: Prof. Haizhou Li) Sept. 2022 – May 2026

- **cGPA:** 3.92 / 4.0 **(Rank**: 1 / 149) **mGPA:** 3.97 / 4.0
- **Honors**: Dean's List, Undergraduate Research Award, Best Undergraduate Teaching Fellow

University of Oxford Oxford, UK
Onsite visiting student, Mathematics and Computer Science Oct. 2024 – Aug. 2025

- **GPA**: 4.0 / 4.0 **(First Class)** Selected as 1 of 30 students for the program
- **Selected Graduate-Level Modules**: Graph Representation Learning A+, Optimal Control A, Uncertainty in Deep Learning A+, Deep Reinforcement Learning A+, Geometric Deep Learning A+, Theories of Deep Learning A+

RESEARCH INTERESTS

My primary research goal is to develop trustworthy and generalizable robotic systems. I am particularly interested in Multimodal Robot Learning and Human-Centered AI (vision-language-action models, attention mechanisms for embodied agents, dexterous manipulation with foundation models, perception-reasoning loops for intent understanding).

RESEARCH EXPERIENCE

Microsoft Research Asia | Machine Learning Group July 2025 – Jan. 2026
Built, trained, and tested VLA models in sim & real settings **2025 IROS Challenge Top 2, Rising Tech Talent**

- Pretrained π0.5-based 3D Vision-Language-Action policies and fine-tuned across 10 dexterous manipulation tasks.
- Introduced a minimum-jerk-inspired regularization term in training to improve motion smoothness and dynamic feasibility.
- Implemented 3D knowledge distillation from geometric teacher models into the visual backbone to capture spatial structure.
- Optimized action chunk size, gripper threshold, and inference frequency on the G1 robot for task-specific control.

Oxford Robotics Institute | Advised by Prof. Shimon Whiteson Oct. 2024 – July 2025
Generalist robot policy learning via vision-language-action models

- Visualized attention maps of VLA policies on images to analyze correlation between layer-wise attention and performance.
- Compared language-only, image-only, and multimodal goals on SimplerEnv and LIBERO benchmarks for generalization.
- Trained an auxiliary vision–language module to predict future image goals from task descriptions and initial observations.
- Implemented Differential Transformer layers in the OTTER backbone on OXE dataset to mitigate late-layer over-smoothing.
- Achieved **+5.3% average success rate** across 5 tasks in SimplerEnv compared to baseline; In submission to ICPR 2026.

Shenzhen Research Institute of Big Data | Advised by Prof. Haizhou Li Sept. 2023 – Aug. 2024
Human multimodal attention & reasoning modeling **2024 ICSR Best Student Paper Award**

- Designed control experiments to study how aligned visual cues affect speech comprehension in multi-speaker environments.
- Analyzed eye-tracking data to quantify multimodal attention patterns; co-authored and presented the paper at ICSR 2024.
- Used large language models to generate structured diagnostic rule sets for specific diseases from clinical text and guidelines
- Developed a sparse-transformer model with two-stage reinforcement learning to capture doctors' perception-reasoning loop.
- Achieved **~45% accuracy** in predicting diagnostic decisions in MIMIC dataset; work under submission to ICML 2026.

PUBLICATIONS

[1] Lichuan Jiang, Jiani Zhong, Muqing Jian, **Xuanzhuo Liu**, Siqi Cai, Haizhou Li, (co-author)
"The Impact of Synchronized Visual and Auditory Attention on Human Perception", **Best Student Paper Award**,
the 16th International Conference on Social Robotics, ICSR 2024, DOI: 10.1007/978-981-96-1151-5_5
[2] **Xuanzhuo Liu**, Zheng Xiong, Shimon Whiteson, under review,
"Differential VLA: Reducing Attention Noise for Robust Vision-Language Action Policy Learning".

TUTORING EXPERIENCE

Undergraduate Teaching Assistant | *CSC3100, STA4001, DDA3020* Feb. 2024 – Present
Technical Volunteer | *Tencent Spark Plan Challenge (for high school students interested in robotics)* May 2025 – July 2025
Voluntary Lecturer | *Oxford Women in CS Society, ML Courses for Non-CS Students* April 2025 – June 2025

HONORS & AWARDS

- National Scholarship Nomination (China's most prestigious undergraduate honor) 2023-2024
- SDS Academic Scholarship (Top 1% academic performance at School of Data Science) 2023-2024
- Diligentia College Residential Scholarships (leadership & community service) 2022-2023
- 2024 Bremen Big Data International Competition | Top 3, Presented at Award Ceremony Dec. 2023